

Paper Summary: Temme et al. 2017 — Zero Noise Extrapolation

Full Citation: Temme, K., Bravyi, S., & Gambetta, J. M. (2017). *Error mitigation for short-depth quantum circuits*. Physical Review Letters, **119**(18), 180509. DOI: 10.1103/PhysRevLett.119.180509. arXiv: 1612.02058.

Problem Statement

Near-term quantum devices ("NISQ") have significant gate errors and decoherence. Without error correction, every gate introduces noise:

$$E_{\lambda}(\rho) = (1 - \lambda)\rho + (\lambda/3) \cdot [X\rho X + Y\rho Y + Z\rho Z]$$

This corrupts expectation values of observables O :

$$\langle O \rangle_{\lambda} = \langle O \rangle_0 + \lambda \epsilon_1 + \lambda^2 \epsilon_2 + O(\lambda^3)$$

Goal: Estimate the noise-free value $\langle O \rangle_0$ without additional qubit resources.

ZNE Method — Richardson Extrapolation

Step 1: Noise Amplification via Gate Folding

Replace each gate U with a folded sequence that has identical ideal action but amplified noise:

$$U \rightarrow U (U^\dagger U)^n \quad (\text{noise scaling factor } c = 2n + 1)$$

This gives effective noise rates $\{c_1\lambda, c_2\lambda, c_3\lambda, \dots\}$ for scale factors $\{c_1, c_2, c_3, \dots\} = \{1, 3, 5, \dots\}$.

Step 2: Measure Expectation Values

Execute circuit at each noise level and record $\langle O \rangle_{\{c_k \lambda\}}$.

Step 3: Richardson Extrapolation

Compute the zero-noise estimate using Richardson's deferred approach to the limit:

$$\langle O \rangle_0 \approx \sum_{k=1}^n c_k \cdot \langle O \rangle_{\lambda_k}$$

where the Richardson coefficients are:

$$c_k = \prod_{j \neq k} [\lambda_j / (\lambda_j - \lambda_k)]$$

Key property: $\sum_k c_k = 1$, and this cancels all noise terms up to order $\lambda^{(n-1)}$.

Key Assumptions

- 1. **Small noise** ($\lambda \ll 1$): Taylor expansion is valid. Method degrades at large λ .
- 2. **Markovian noise**: Noise is uncorrelated between gates.
- 3. **Gate-independent noise**: Same λ applies to every gate.
- 4. **Locality**: Single-qubit depolarizing after each gate.

What We Reproduce

Figure	Original Result	Our Result	Status
Fig 1: $\langle ZZ \rangle$ vs λ	Linear decay $\approx (1 - 4\lambda/3)^d$	✅ Linear decay confirmed	Reproduced
Fig 2: ZNE vs raw error	ZNE reduces error $\sim 5\text{-}10\times$ at low λ	✅ Confirmed at $\lambda \leq 0.02$	Reproduced
Fig 3: Method comparison	Richardson outperforms linear 2-point	✅ Richardson has lower error	Reproduced

See `DISCREPANCY_NOTES.md` for documented differences.